2.1.1 Land Surface Temperature and Surface Urban Heat Islands

Your discussion here must establish the physical foundations of thermal remote sensing, justify using an urban-rural contrast rather than absolute temperature, and set clear conceptual boundaries regarding pedestrian heat exposure.

- **Li et al. (2023)**
  - **What to find & discuss:** The physical definition of Land Surface Temperature (LST) as radiometric skin temperature calculated from top-of-atmosphere thermal infrared (TIR) radiance. Note its retrieval as an "ill-posed" inverse problem requiring atmospheric correction and surface emissivity estimation.
  - **Thesis tie-in:** Establishes the physical definition of your target variable (y) as radiometric surface skin temperature rather than ambient air temperature.
- **Zhou et al. (2019)**
  - **What to find & discuss:** The formal definition of SUHI intensity ($\Delta LST = LST_{\text{urban}} - LST_{\text{non-urban}}$). Look for their discussion on the discrepancies caused by diurnal timing, satellite overpass times, and how non-urban reference definitions vary across literature.
  - **Thesis tie-in:** Justifies why high LST alone does not constitute an urban heat island and why you must evaluate a differential intensity metric.
- **Yang & Yao (2022)**
  - **What to find & discuss:** Their empirical analysis of 346 global cities showing that SUHI intensity fluctuates drastically based on the vegetation composition of the rural baseline. Find the finding that when reference land cover varies, SUHI appears correlated with precipitation, but fixing reference land cover stabilizes the metric.
  - **Thesis tie-in:** The theoretical defense for your methodology fixing an explicit non-urban reference mask before running model evaluations.
- **Guo et al. (2022)**
  - **What to find & discuss:** Their urban-rural contrast method in Lagos, West Africa. Extract their metric showing that localized urbanization accounted for ~61% of urban surface warming compared to macro-climatic trends.
  - **Thesis tie-in:** Proves that subtracting a non-urban rural baseline isolates urban infrastructure heating from regional background climate change.
- **Liu et al. (2022)**
  - **What to find & discuss:** Findings from 2,000+ global city clusters showing that urban cores experience surface warming rates 29% faster than their surrounding rural backgrounds ($0.50 \pm 0.20\text{ K/decade}$).
  - **Thesis tie-in:** Empirical evidence of divergence between urban built-up surfaces and rural baselines over multi-decadal timeframes.
- **Zhu et al. (2025)**
  - **What to find & discuss:** Thermal gradients across 19 Southeast Asian cities. Note their finding that larger cities bleed heat outwards into peri-urban and rural areas.
  - **Thesis tie-in:** Explains why your non-urban reference buffer must be

placed far enough from the urban edge to prevent contamination from peri-urban thermal expansion.

- **Venter et al. (2021)**
  - **What to find & discuss:** The empirical comparison of >50,000 crowdsourced weather stations with satellite LST across 342 European cities. Extract the exact statistic: satellites produce a **sixfold overestimate** of UHI intensity compared to canopy air measurements (1.45°C vs. 0.26°C).
  - **Thesis tie-in:** Your primary justification that satellite LST models estimate physical ground surface skin temperature, not human heat exposure, ambient air temperature, or health risk.
- **Delina et al. (2026)**
  - **What to find & discuss:** Their findings in Metro Manila and Bangkok that satellite LST clusters miss pedestrian microhotspots and lived coping practices.
  - **Thesis tie-in:** Reinforces that your study strictly limits its scope to environmental surface estimation, explicitly excluding pedestrian heat exposure or health assessments.

2.1.2 Landsat-Based Environmental Observations

Your discussion here must validate the satellite data products, justify annual compositing and cloud masking, explain spectral indices, and outline mixed-pixel limits along road vectors.

- **Ayek et al. (2025)**
  - **What to find & discuss:** The ground validation of Landsat-8 Collection 2 Level-2 (C2L2) LST against SURFRAD stations. Note the overall accuracy ($R^2 = 0.91$, $r = 0.98$), but specifically extract the limitations: a systematic warm bias of $+1.75^\circ\text{C}$ and increased retrieval errors during high surface temperatures ($>40^\circ\text{C}$) and high atmospheric water vapor.
  - **Thesis tie-in:** Validates your primary data source while directly acknowledging sensor biases in humid tropical environments like Davao City.
- **Ermida et al. (2020)**
  - **What to find & discuss:** The technical framework of using USGS Landsat Collection Tier 1 data (radiometrically calibrated for multi-year time series) and pixel-level quality assessment (QA/BQA) band masking to filter clouds, cirrus, and cloud shadows.
  - **Thesis tie-in:** Methodological precedent for your automated cloud-masking and data-cleaning pipeline.
- **Gorelick et al. (2017) & Amani et al. (2020)**
  - **What to find & discuss:** How Google Earth Engine executes temporal pixel reducers. Find the rationale for **annual median compositing**: it statistically filters out unmasked transient cloud artifacts, sensor dropouts, and shadow anomalies across multi-temporal stacks.
  - **Thesis tie-in:** Defends your operational workflow of computing annual median composites (2019–2025) rather than relying on noisy single-date scenes.
- **Zha, Gao & Ni (2003)**
  - **What to find & discuss:** The mathematical development of the

Normalized Difference Built-up Index ($\text{NDBI} = \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}}$). Extract the physical principle that built surfaces reflect higher in SWIR than NIR.
    - **Thesis tie-in:** Provides foundational documentation for NDBI while supporting your scoping caveat that NDBI is an automated spectral proxy, not a direct physical measurement of impervious surface density.
- **Weng, Lu & Schubring (2004)**
    - **What to find & discuss:** The negative correlation between NDVI (vegetation abundance) and LST, and how their spatial relationship scales across pixel resolutions (30 m to 120 m).
    - **Thesis tie-in:** Justifies using NDVI as a baseline vegetation feature in thermal regression.
- **Andriambololonaharisoamalala et al. (2024)**
    - **What to find & discuss:** The systematic review of spatial resolution limits in thermal remote sensing. Extract the concept of the **blending effect (mixed pixels)**: thermal infrared sensors collect radiant emissions over a 30–100 m footprint, combining asphalt roads, adjacent buildings, and roadside trees into a single value.
    - **Thesis tie-in:** Explicitly defends your delimitation that placing satellite estimates onto densified road nodes does not create true micro-scale pavement or street-canyon measurements.

2.1.3 Urban Surface-Temperature Research in Tropical and Philippine Cities

Your discussion here must establish the regional climatic setting, review Philippine studies linking spectral indices to LST, detail local Davao City thermal dynamics, and state the research gap.

- **Nguyen et al. (2022) & Ng et al. (2025)**
    - **What to find & discuss:** Multi-decadal satellite trends in tropical Southeast Asian cities (e.g., Bangkok, Ho Chi Minh, Kuala Lumpur). Extract the warming rates (e.g., 0.215°C to $0.235^\circ\text{C}/\text{year}$) and the conclusion that rapid impervious expansion paired with vegetative loss causes persistent thermal intensification in equatorial climates.
    - **Thesis tie-in:** Establishes the macro Southeast Asian regional context for your tropical study area.
- **Tun et al. (2025)**
    - **What to find & discuss:** Review of cooling strategies in Southeast Asia. Note how monsoonal humidity and dense urban geometries reduce natural convective cooling, whereas roadside street-tree networks provide up to 11°C in surface temperature reduction.
    - **Thesis tie-in:** Connects roadside environmental characteristics directly to surface temperature dynamics.
- **Manalo et al. (2022)**
    - **What to find & discuss:** Analysis of 34 Philippine meteorological stations (1951–2018). Extract their finding that urbanization significantly accelerates minimum nighttime temperatures (T_{min}) over daytime temperatures, shrinking the diurnal

temperature range, with the strongest urban-rural difference occurring during the cool dry season (DJF).
    - **Thesis tie-in:** Establishes the national atmospheric context of urbanization in the Philippines while reiterating that meteorological station data must not be conflated with satellite SUHI skin temperatures.
- **Galvez et al. (2024)**
    - **What to find & discuss:** Land cover and LST in Cagayan de Oro River basin, Mindanao (2003–2020). Extract their finding that mean LST increased by 5.48°C as croplands and built-up areas expanded, with built surfaces consistently recording the highest temperatures.
    - **Thesis tie-in:** Documents regional surface thermal dynamics specific to Mindanao.
- **Almadrones-Reyes & Dagamac (2023) & Purio et al. (2022)**
    - **What to find & discuss:** Landsat time-series studies in Metro Manila. Note the 4.02°C warming linked to rising NDBI and falling NDVI, and Purio's observation that asphalt/concrete roads and transport corridors form the warmest contiguous zones.
    - **Thesis tie-in:** Validates the predictive link between built-up indices, road infrastructure, and elevated LST in the Philippines.
- **Treyes (2025)**
    - **What to find & discuss:** Intra-urban heat islets covering 31% of Quezon City, demonstrating spatial autocorrelation and localized clustering.
    - **Thesis tie-in:** Confirms that urban thermal variation in Philippine cities is non-random and spatially clustered.

Local Davao City Research & The Unresolved Gap

- **Tinoy et al. (2019)**
    - **What to find & discuss:** Multiple regression and quantile regression of LST in Davao City (1994–2019). Extract their key finding: **NDBI is the single most significant predictor of LST at the 50th, 75th, and 90th quantiles** (31.87°C to 33.42°C) in core districts (Poblacion, Talomo, Agdao, Buhangin).
- **Batican & Paragamac (2025)**
    - **What to find & discuss:** Spatiotemporal analysis across Davao City's three congressional districts (2015–2024). Extract their finding that District I barangays (1-A, 7-A, 11-B) experience peak surface temperatures approaching or exceeding 38°C.
- **Dumdum & Paragamac (2026)**
    - **What to find & discuss:** Landsat 8 analysis in Davao City (2018–2022). Extract their correlation statistics: NDBI vs. LST ($r = 0.6216$), NDVI vs. LST ($r = -0.2495$), and their specific finding that major arterial corridors (Matina and Airport road) exhibit high thermal concentration, leading to recommendations for roadside green buffers.
- **The Critical Gap to Conclude 2.1.3:**
    - **Discussion to write:** Synthesize that while Tinoy et al., Batican & Paragamac, and Dumdum & Paragamac confirm that built infrastructure elevates surface temperatures across Davao City, all

existing studies treat locations as **isolated pixels or administrative polygons** using global regression models. None investigate whether the **network topology and connectivity of the road system itself** contribute useful predictive information for estimating surface temperatures along urban transit infrastructure. This gap directly motivates your proposed graph model.